

```
/* USER CODE BEGIN Header /
```

```
/*
```

- @file : main.c
- @brief : Main program body

- @attention
-
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-
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-

```
/
```

```
/USER CODE END Header /
```

```
/ Includes -----*/
```

```
#include "main.h"
```

```
/* Private includes -----/
```

```
/USER CODE BEGIN Includes /
```

```
#include <stdio.h>
```

```
#include "stm32f4xx.h"
```

```
#include "lcd_stm32f4.h"
```

```
/USER CODE END Includes */
```

```
/* Private typedef -----/
```

```
/USER CODE BEGIN PTD */
```

```
/* USER CODE END PTD */
```

```
/* Private define -----/
```

```
/USER CODE BEGIN PD */
```

```
// TODO: Add values for below variables
```

```
#define NS 1024 // Number of samples in LUT
```

```
#define TIM2CLK 16000000 // STM Clock frequency: Hint You might want to check the ioc file
```

```
#define F_SIGNAL 0.5 // Frequency of output analog signal
```

```
/* USER CODE END PD */
```

```
/* Private macro -----/
```

```
/USER CODE BEGIN PM */
```

```
/* USER CODE END PM */
```

```
/* Private variables -----*/
```

```
TIM_HandleTypeDef htim2;
```

```
TIM_HandleTypeDef htim3;
```

```
DMA_HandleTypeDef hdma_tim2_ch1;
```

```
/* USER CODE BEGIN PV */
```

```
// TODO: Add code for global variables, including LUTs
```

```
const uint32_t Sin_LUT[NS] =
```

```
{2048,2060,2073,2085,2098,2110,2123,2135,2148,2161,2173,2186,2198,2211,2223,2236,2248,2261,2273,2286,229
8,2311,2323,2335,2348,2360,2373,2385,2398,2410,2422,2435,2447,2459,2472,2484,2496,2508,2521,2533,2545,25
57,2569,2581,2594,2606,2618,2630,2642,2654,2666,2678,2690,2702,2714,2725,2737,2749,2761,2773,2784,2796,2
808,2819,2831,2843,2854,2866,2877,2889,2900,2912,2923,2934,2946,2957,2968,2979,2990,3002,3013,3024,3035,
3046,3057,3068,3078,3089,3100,3111,3122,3132,3143,3154,3164,3175,3185,3195,3206,3216,3226,3237,3247,3257
,3267,3277,3287,3297,3307,3317,3327,3337,3346,3356,3366,3375,3385,3394,3404,3413,3423,3432,3441,3450,345
9,3468,3477,3486,3495,3504,3513,3522,3530,3539,3548,3556,3565,3573,3581,3590,3598,3606,3614,3622,3630,36
38,3646,3654,3662,3669,3677,3685,3692,3700,3707,3714,3722,3729,3736,3743,3750,3757,3764,3771,3777,3784,3
791,3797,3804,3810,3816,3823,3829,3835,3841,3847,3853,3859,3865,3871,3876,3882,3888,3893,3898,3904,3909,
3914,3919,3924,3929,3934,3939,3944,3949,3953,3958,3962,3967,3971,3975,3980,3984,3988,3992,3996,3999,400
3,4007,4010,4014,4017,4021,4024,4027,4031,4034,4037,4040,4042,4045,4048,4051,4053,4056,4058,4060,4063,40
65,4067,4069,4071,4073,4075,4076,4078,4080,4081,4083,4084,4085,4086,4087,4088,4089,4090,4091,4092,4093,4
093,4094,4094,4094,4095,4095,4095,4095,4095,4095,4095,4095,4094,4094,4094,4093,4093,4092,4091,4090,4089,4088,
4087,4086,4085,4084,4083,4081,4080,4078,4076,4075,4073,4071,4069,4067,4065,4063,4060,4058,4056,4053,405
1,4048,4045,4042,4040,4037,4034,4031,4027,4024,4021,4017,4014,4010,4007,4003,3999,3996,3992,3988,3984,39
80,3975,3971,3967,3962,3958,3953,3949,3944,3939,3934,3929,3924,3919,3914,3909,3904,3898,3893,3888,3882,3
876,3871,3865,3859,3853,3847,3841,3835,3829,3823,3816,3810,3804,3797,3791,3784,3777,3771,3764,3757,3750,
3743,3736,3729,3722,3714,3707,3700,3692,3685,3677,3669,3662,3654,3646,3638,3630,3622,3614,3606,3598,359
0,3581,3573,3565,3556,3548,3539,3530,3522,3513,3504,3495,3486,3477,3468,3459,3450,3441,3432,3423,3413,34
04,3394,3385,3375,3366,3356,3346,3337,3327,3317,3307,3297,3287,3277,3267,3257,3247,3237,3226,3216,3206,3
195,3185,3175,3164,3154,3143,3132,3122,3111,3100,3089,3078,3068,3057,3046,3035,3024,3013,3002,2990,2979,
2968,2957,2946,2934,2923,2912,2900,2889,2877,2866,2854,2843,2831,2819,2808,2796,2784,2773,2761,2749,273
7,2725,2714,2702,2690,2678,2666,2654,2642,2630,2618,2606,2594,2581,2569,2557,2545,2533,2521,2508,2496,24
84,2472,2459,2447,2435,2422,2410,2398,2385,2373,2360,2348,2335,2323,2311,2298,2286,2273,2261,2248,2236,2
223,2211,2198,2186,2173,2161,2148,2135,2123,2110,2098,2085,2073,2060,2048,2035,2022,2010,1997,1985,1972,
1960,1947,1934,1922,1909,1897,1884,1872,1859,1847,1834,1822,1809,1797,1784,1772,1760,1747,1735,1722,171
0,1697,1685,1673,1660,1648,1636,1623,1611,1599,1587,1574,1562,1550,1538,1526,1514,1501,1489,1477,1465,14
53,1441,1429,1417,1405,1393,1381,1370,1358,1346,1334,1322,1311,1299,1287,1276,1264,1252,1241,1229,1218,1
206,1195,1183,1172,1161,1149,1138,1127,1116,1105,1093,1082,1071,1060,1049,1038,1027,1017,1006,995,984,973,
963,952,941,931,920,910,900,889,879,869,858,848,838,828,818,808,798,788,778,768,758,749,739,729,720,710,70
1,691,682,672,663,654,645,636,627,618,609,600,591,582,573,565,556,547,539,530,522,514,505,497,489,481,473,4
65,457,449,441,433,426,418,410,403,395,388,381,373,366,359,352,345,338,331,324,318,311,304,298,291,285,279,
272,266,260,254,248,242,236,230,224,219,213,207,202,197,191,186,181,176,171,166,161,156,151,146,142,137,13
3,128,124,120,115,111,107,103,99,96,92,88,85,81,78,74,71,68,64,61,58,55,53,50,47,44,42,39,37,35,32,30,28,26,24,
22,20,19,17,15,14,12,11,10,9,8,7,6,5,4,3,2,2,1,1,1,0,0,0,0,0,0,1,1,1,2,2,3,4,5,6,7,8,9,10,11,12,14,15,17,19,20,22,24,
26,28,30,32,35,37,39,42,44,47,50,53,55,58,61,64,68,71,74,78,81,85,88,92,96,99,103,107,111,115,120,124,128,133,1
37,142,146,151,156,161,166,171,176,181,186,191,197,202,207,213,219,224,230,236,242,248,254,260,266,272,279,
285,291,298,304,311,318,324,331,338,345,352,359,366,373,381,388,395,403,410,418,426,433,441,449,457,465,47
```

3,481,489,497,505,514,522,530,539,547,556,565,573,582,591,600,609,618,627,636,645,654,663,672,682,691,701,710,720,729,739,749,758,768,778,788,798,808,818,828,838,848,858,869,879,889,900,910,920,931,941,952,963,973,984,995,1006,1017,1027,1038,1049,1060,1071,1082,1093,1105,1116,1127,1138,1149,1161,1172,1183,1195,1206,1218,1229,1241,1252,1264,1276,1287,1299,1311,1322,1334,1346,1358,1370,1381,1393,1405,1417,1429,1441,1453,1465,1477,1489,1501,1514,1526,1538,1550,1562,1574,1587,1599,1611,1623,1636,1648,1660,1673,1685,1697,1710,1722,1735,1747,1760,1772,1784,1797,1809,1822,1834,1847,1859,1872,1884,1897,1909,1922,1934,1947,1960,1972,1985,1997,2010,2022,2035};

const uint32_t Saw_LUT[NS] =

{0,4,8,12,16,20,24,28,32,36,40,44,48,52,56,60,64,68,72,76,80,84,88,92,96,100,104,108,112,116,120,124,128,132,136,140,144,148,152,156,160,164,168,172,176,180,184,188,192,196,200,204,208,212,216,220,224,228,232,236,240,244,248,252,256,260,264,268,272,276,280,284,288,292,296,300,304,308,312,316,320,324,328,332,336,340,344,348,352,356,360,364,368,372,376,380,384,388,392,396,400,404,408,412,416,420,424,428,432,436,440,444,448,452,456,460,464,468,472,476,480,484,488,492,496,500,504,508,512,516,520,524,528,532,536,540,544,548,552,556,560,564,568,572,576,580,584,588,592,596,600,604,608,612,616,620,624,628,632,636,640,644,648,652,656,660,664,668,672,676,680,684,688,692,696,700,704,708,712,716,720,724,728,732,736,740,744,748,752,756,760,764,768,772,776,780,784,788,792,796,800,804,808,812,816,820,824,828,832,836,840,844,848,852,856,860,864,868,872,876,880,884,888,892,896,900,904,908,912,916,920,924,928,932,936,940,944,948,952,956,960,964,968,972,976,980,984,988,992,996,1000,1004,1008,1012,1016,1020,1024,1028,1032,1036,1040,1044,1048,1052,1056,1060,1064,1068,1072,1076,1080,1084,1088,1092,1096,1100,1104,1108,1112,1116,1120,1124,1128,1132,1136,1140,1144,1148,1152,1156,1160,1164,1168,1172,1176,1180,1184,1188,1192,1196,1200,1204,1208,1212,1216,1220,1224,1228,1232,1236,1240,1244,1248,1252,1256,1260,1264,1268,1272,1276,1280,1284,1288,1292,1296,1300,1304,1308,1312,1316,1320,1324,1328,1332,1336,1340,1344,1348,1352,1356,1360,1364,1368,1372,1376,1380,1384,1388,1392,1396,1400,1404,1408,1412,1416,1420,1424,1428,1432,1436,1440,1444,1448,1452,1456,1460,1464,1468,1472,1476,1480,1484,1488,1492,1496,1500,1504,1508,1512,1516,1520,1524,1528,1532,1536,1540,1544,1548,1552,1556,1560,1564,1568,1572,1576,1580,1584,1588,1592,1596,1600,1604,1608,1612,1616,1620,1624,1628,1632,1636,1640,1644,1648,1652,1656,1660,1664,1668,1672,1676,1680,1684,1688,1692,1696,1700,1704,1708,1712,1716,1720,1724,1728,1732,1736,1740,1744,1748,1752,1756,1760,1764,1768,1772,1776,1780,1784,1788,1792,1796,1800,1804,1808,1812,1816,1820,1824,1828,1832,1836,1840,1844,1848,1852,1856,1860,1864,1868,1872,1876,1880,1884,1888,1892,1896,1900,1904,1908,1912,1916,1920,1924,1928,1932,1936,1940,1944,1948,1952,1956,1960,1964,1968,1972,1976,1980,1984,1988,1992,1996,2000,2004,2008,2012,2016,2020,2024,2028,2032,2036,2040,2044,2048,2052,2056,2060,2064,2068,2072,2076,2080,2084,2088,2092,2096,2099,2103,2107,2111,2115,2119,2123,2127,2131,2135,2139,2143,2147,2151,2155,2159,2163,2167,2171,2175,2179,2183,2187,2191,2195,2199,2203,2207,2211,2215,2219,2223,2227,2231,2235,2239,2243,2247,2251,2255,2259,2263,2267,2271,2275,2279,2283,2287,2291,2295,2299,2303,2307,2311,2315,2319,2323,2327,2331,2335,2339,2343,2347,2351,2355,2359,2363,2367,2371,2375,2379,2383,2387,2391,2395,2399,2403,2407,2411,2415,2419,2423,2427,2431,2435,2439,2443,2447,2451,2455,2459,2463,2467,2471,2475,2479,2483,2487,2491,2495,2499,2503,2507,2511,2515,2519,2523,2527,2531,2535,2539,2543,2547,2551,2555,2559,2563,2567,2571,2575,2579,2583,2587,2591,2595,2599,2603,2607,2611,2615,2619,2623,2627,2631,2635,2639,2643,2647,2651,2655,2659,2663,2667,2671,2675,2679,2683,2687,2691,2695,2699,2703,2707,2711,2715,2719,2723,2727,2731,2735,2739,2743,2747,2751,2755,2759,2763,2767,2771,2775,2779,2783,2787,2791,2795,2799,2803,2807,2811,2815,2819,2823,2827,2831,2835,2839,2843,2847,2851,2855,2859,2863,2867,2871,2875,2879,2883,2887,2891,2895,2899,2903,2907,2911,2915,2919,2923,2927,2931,2935,2939,2943,2947,2951,2955,2959,2963,2967,2971,2975,2979,2983,2987,2991,2995,2999,3003,3007,3011,3015,3019,3023,3027,3031,3035,3039,3043,3047,3051,3055,3059,3063,3067,3071,3075,3079,3083,3087,3091,3095,3099,3103,3107,3111,3115,3119,3123,3127,3131,3135,3139,3143,3147,3151,3155,3159,3163,3167,3171,3175,3179,3183,3187,3191,3195,3199,3203,3207,3211,3215,3219,3223,3227,3231,3235,3239,3243,3247,3251,3255,3259,3263,3267,3271,3275,3279,3283,3287,3291,3295,3299,3303,3307,3311,3315,3319,3323,3327,3331,3335,3339,3343,3347,3351,3355,3359,3363,3367,3371,3375,3379,3383,3387,3391,3395,3399,3403,3407,3411,3415,3419,3423,3427,3431,3435,3439,3443,3447,3451,3455,3459,3463,3467,3471,3475,3479,3483,

3487,3491,3495,3499,3503,3507,3511,3515,3519,3523,3527,3531,3535,3539,3543,3547,3551,3555,3559,3563,3567,3571,3575,3579,3583,3587,3591,3595,3599,3603,3607,3611,3615,3619,3623,3627,3631,3635,3639,3643,3647,3651,3655,3659,3663,3667,3671,3675,3679,3683,3687,3691,3695,3699,3703,3707,3711,3715,3719,3723,3727,3731,3735,3739,3743,3747,3751,3755,3759,3763,3767,3771,3775,3779,3783,3787,3791,3795,3799,3803,3807,3811,3815,3819,3823,3827,3831,3835,3839,3843,3847,3851,3855,3859,3863,3867,3871,3875,3879,3883,3887,3891,3895,3899,3903,3907,3911,3915,3919,3923,3927,3931,3935,3939,3943,3947,3951,3955,3959,3963,3967,3971,3975,3979,3983,3987,3991,3995,3999,4003,4007,4011,4015,4019,4023,4027,4031,4035,4039,4043,4047,4051,4055,4059,4063,4067,4071,4075,4079,4083,4087,4091};

const uint32_t Triangle_LUT[NS] =

{0,8,16,24,32,40,48,56,64,72,80,88,96,104,112,120,128,136,144,152,160,168,176,184,192,200,208,216,224,232,240,248,256,264,272,280,288,296,304,312,320,328,336,344,352,360,368,376,384,392,400,408,416,424,432,440,448,456,464,472,480,488,496,504,512,520,528,536,544,552,560,568,576,584,592,600,608,616,624,632,640,648,656,664,672,680,688,696,704,712,720,728,736,744,752,760,768,776,784,792,800,808,816,824,832,840,848,856,864,872,880,888,896,904,912,920,928,936,944,952,960,968,976,984,992,1000,1008,1016,1024,1032,1040,1048,1056,1064,1072,1080,1088,1096,1104,1112,1120,1128,1136,1144,1152,1160,1168,1176,1184,1192,1200,1208,1216,1224,1232,1240,1248,1256,1264,1272,1280,1288,1296,1304,1312,1320,1328,1336,1344,1352,1360,1368,1376,1384,1392,1400,1408,1416,1424,1432,1440,1448,1456,1464,1472,1480,1488,1496,1504,1512,1520,1528,1536,1544,1552,1560,1568,1576,1584,1592,1600,1608,1616,1624,1632,1640,1648,1656,1664,1672,1680,1688,1696,1704,1712,1720,1728,1736,1744,1752,1760,1768,1776,1784,1792,1800,1808,1816,1824,1832,1840,1848,1856,1864,1872,1880,1888,1896,1904,1912,1920,1928,1936,1944,1952,1960,1968,1976,1984,1992,2000,2008,2016,2024,2032,2040,2048,2056,2064,2072,2080,2088,2096,2104,2112,2120,2128,2136,2144,2152,2160,2168,2176,2184,2192,2200,2208,2216,2224,2232,2240,2248,2256,2264,2272,2280,2288,2296,2304,2312,2320,2328,2336,2344,2352,2360,2368,2376,2384,2392,2400,2408,2416,2424,2432,2440,2448,2456,2464,2472,2480,2488,2496,2504,2512,2520,2528,2536,2544,2552,2560,2568,2576,2584,2592,2600,2608,2616,2624,2632,2640,2648,2656,2664,2672,2680,2688,2696,2704,2712,2720,2728,2736,2744,2752,2760,2768,2776,2784,2792,2800,2808,2816,2824,2832,2840,2848,2856,2864,2872,2880,2888,2896,2904,2912,2920,2928,2936,2944,2952,2960,2968,2976,2984,2992,3000,3008,3016,3024,3032,3040,3048,3056,3064,3072,3080,3088,3096,3104,3112,3120,3128,3136,3144,3152,3160,3168,3176,3184,3192,3200,3208,3216,3224,3232,3240,3248,3256,3264,3272,3280,3288,3296,3304,3312,3320,3328,3336,3344,3352,3360,3368,3376,3384,3392,3400,3408,3416,3424,3432,3440,3448,3456,3464,3472,3480,3488,3496,3504,3512,3520,3528,3536,3544,3552,3560,3568,3576,3584,3592,3600,3608,3616,3624,3632,3640,3648,3656,3664,3672,3680,3688,3696,3704,3712,3720,3728,3736,3744,3752,3760,3768,3776,3784,3792,3800,3808,3816,3824,3832,3840,3848,3856,3864,3872,3880,3888,3896,3904,3912,3920,3928,3936,3944,3952,3960,3968,3976,3984,3992,4000,4008,4016,4024,4032,4040,4048,4056,4064,4072,4080,4088,4096,4104,4112,4120,4128,4136,4144,4152,4160,4168,4176,4184,4192,4200,4208,4216,4224,4232,4240,4248,4256,4264,4272,4280,4288,4296,4304,4312,4320,4328,4336,4344,4352,4360,4368,4376,4384,4392,4400,4408,4416,4424,4432,4440,4448,4456,4464,4472,4480,4488,4496,4504,4512,4520,4528,4536,4544,4552,4560,4568,4576,4584,4592,4600,4608,4616,4624,4632,4640,4648,4656,4664,4672,4680,4688,4696,4704,4712,4720,4728,4736,4744,4752,4760,4768,4776,4784,4792,4800,4808,4816,4824,4832,4840,4848,4856,4864,4872,4880,4888,4896,4904,4912,4920,4928,4936,4944,4952,4960,4968,4976,4984,4992,5000,5008,5016,5024,5032,5040,5048,5056,5064,5072,5080,5088,5096,5104,5112,5120,5128,5136,5144,5152,5160,5168,5176,5184,5192,5200,5208,5216,5224,5232,5240,5248,5256,5264,5272,5280,5288,5296,5304,5312,5320,5328,5336,5344,5352,5360,5368,5376,5384,5392,5400,5408,5416,5424,5432,5440,5448,5456,5464,5472,5480,5488,5496,5504,5512,5520,5528,5536,5544,5552,5560,5568,5576,5584,5592,5600,5608,5616,5624,5632,5640,5648,5656,5664,5672,5680,5688,5696,5704,5712,5720,5728,5736,5744,5752,5760,5768,5776,5784,5792,5800,5808,5816,5824,5832,5840,5848,5856,5864,5872,5880,5888,5896,5904,5912,5920,5928,5936,5944,5952,5960,5968,5976,5984,5992,6000,6008,6016,6024,6032,6040,6048,6056,6064,6072,6080,6088,6096,6104,6112,6120,6128,6136,6144,6152,6160,6168,6176,6184,6192,6200,6208,6216,6224,6232,6240,6248,6256,6264,6272,6280,6288,6296,6304,6312,6320,6328,6336,6344,6352,6360,6368,6376,6384,6392,6400,6408,6416,6424,6432,6440,6448,6456,6464,6472,6480,6488,6496,6504,6512,6520,6528,6536,6544,6552,6560,6568,6576,6584,6592,6600,6608,6616,6624,6632,6640,6648,6656,6664,6672,6680,6688,6696,6704,6712,6720,6728,6736,6744,6752,6760,6768,6776,6784,6792,6800,6808,6816,6824,6832,6840,6848,6856,6864,6872,6880,6888,6896,6904,6912,6920,6928,6936,6944,6952,6960,6968,6976,6984,6992,7000,7008,7016,7024,7032,7040,7048,7056,7064,7072,7080,7088,7096,7104,7112,7120,7128,7136,7144,7152,7160,7168,7176,7184,7192,7200,7208,7216,7224,7232,7240,7248,7256,7264,7272,7280,7288,7296,7304,7312,7320,7328,7336,7344,7352,7360,7368,7376,7384,7392,7400,7408,7416,7424,7432,7440,7448,7456,7464,7472,7480,7488,7496,7504,7512,7520,7528,7536,7544,7552,7560,7568,7576,7584,7592,7600,7608,7616,7624,7632,7640,7648,7656,7664,7672,7680,7688,7696,7704,7712,7720,7728,7736,7744,7752,7760,7768,7776,7784,7792,7800,7808,7816,7824,7832,7840,7848,7856,7864,7872,7880,7888,7896,7904,7912,7920,7928,7936,7944,7952,7960,7968,7976,7984,7992,8000,8008,8016,8024,8032,8040,8048,8056,8064,8072,8080,8088,8096,8104,8112,8120,8128,8136,8144,8152,8160,8168,8176,8184,8192,8200,8208,8216,8224,8232,8240,8248,8256,8264,8272,8280,8288,8296,8304,8312,8320,8328,8336,8344,8352,8360,8368,8376,8384,8392,8400,8408,8416,8424,8432,8440,8448,8456,8464,8472,8480,8488,8496,8504,8512,8520,8528,8536,8544,8552,8560,8568,8576,8584,8592,8600,8608,8616,8624,8632,8640,8648,8656,8664,8672,8680,8688,8696,8704,8712,8720,8728,8736,8744,8752,8760,8768,8776,8784,8792,8800,8808,8816,8824,8832,8840,8848,8856,8864,8872,8880,8888,8896,8904,8912,8920,8928,8936,8944,8952,8960,8968,8976,8984,8992,9000,9008,9016,9024,9032,9040,9048,9056,9064,9072,9080,9088,9096,9104,9112,9120,9128,9136,9144,9152,9160,9168,9176,9184,9192,9200,9208,9216,9224,9232,9240,9248,9256,9264,9272,9280,9288,9296,9304,9312,9320,9328,9336,9344,9352,9360,9368,9376,9384,9392,9400,9408,9416,9424,9432,9440,9448,9456,9464,9472,9480,9488,9496,9504,9512,9520,9528,9536,9544,9552,9560,9568,9576,9584,9592,9600,9608,9616,9624,9632,9640,9648,9656,9664,9672,9680,9688,9696,9704,9712,9720,9728,9736,9744,9752,9760,9768,9776,9784,9792,9800,9808,9816,9824,9832,9840,9848,9856,9864,9872,9880,9888,9896,9904,9912,9920,9928,9936,9944,9952,9960,9968,9976,9984,9992,10000,10008,10016,10024,10032,10040,10048,10056,10064,10072,10080,10088,10096,10104,10112,10120,10128,10136,10144,10152,10160,10168,10176,10184,10192,10200,10208,10216,10224,10232,10240,10248,10256,10264,10272,10280,10288,10296,10304,10312,10320,10328,10336,10344,10352,10360,10368,10376,10384,10392,10400,10408,10416,10424,10432,10440,10448,10456,10464,10472,10480,10488,10496,10504,10512,10520,10528,10536,10544,10552,10560,10568,10576,10584,10592,10600,10608,10616,10624,10632,10640,10648,10656,10664,10672,10680,10688,10696,10704,10712,10720,10728,10736,10744,10752,10760,10768,10776,10784,10792,10800,10808,10816,10824,10832,10840,10848,10856,10864,10872,10880,10888,10896,10904,10912,10920,10928,10936,10944,10952,10960,10968,10976,10984,10992,11000,11008,11016,11024,11032,11040,11048,11056,11064,11072,11080,11088,11096,11104,11112,11120,11128,11136,11144,11152,11160,11168,11176,11184,11192,11200,11208,11216,11224,11232,11240,11248,11256,11264,11272,11280,11288,11296,11304,11312,11320,11328,11336,11344,11352,11360,11368,11376,11384,11392,11400,11408,11416,11424,11432,11440,11448,11456,11464,11472,11480,11488,11496,11504,11512,11520,11528,11536,11544,11552,11560,11568,11576,11584,11592,11600,11608,11616,11624,11632,11640,11648,11656,11664,11672,11680,11688,11696,11704,11712,11720,11728,11736,11744,11752,11760,11768,11776,11784,11792,11800,11808,11816,11824,11832,11840,11848,11856,11864,11872,11880,11888,11896,11904,11912,11920,11928,11936,11944,11952,11960,11968,11976,11984,11992,12000,12008,12016,12024,12032,12040,12048,12056,12064,12072,12080,12088,12096,12104,12112,12120,12128,12136,12144,12152,12160,12168,12176,12184,12192,12200,12208,12216,12224,12232,12240,12248,12256,12264,12272,12280,12288,12296,12304,12312,12320,12328,12336,12344,12352,12360,12368,12376,12384,12392,12400,12408,12416,12424,12432,12440,12448,12456,12464,12472,12480,12488,12496,12504,12512,12520,12528,12536,12544,12552,12560,12568,12576,12584,12592,12600,12608,12616,12624,12632,12640,12648,12656,12664,12672,12680,12688,12696,12704,12712,12720,12728,12736,12744,12752,12760,12768,12776,12784,12792,12800,12808,12816,12824,12832,12840,12848,12856,12864,12872,12880,12888,12896,12904,12912,12920,12928,12936,12944,12952,12960,12968,12976,12984,12992,13000,13008,13016,13024,13032,13040,13048,13056,13064,13072,13080,13088,13096,13104,13112,13120,13128,13136,13144,13152,13160,13168,13176,13184,13192,13200,13208,13216,13224,13232,13240,13248,13256,13264,13272,13280,13288,13296,13304,13312,13320,13328,13336,13344,13352,13360,13368,13376,13384,13392,13400,13408,13416,13424,13432,13440,13448,13456,13464,13472,13480,13488,13496,13504,13512,13520,13528,13536,13544,13552,13560,13568,13576,13584,13592,13600,13608,13616,13624,13632,13640,13648,13656,13664,13672,13680,13688,13696,13704,13712,13720,13728,13736,13744,13752,13760,13768,13776,13784,13792,13800,13808,13816,13824,13832,13840,13848,13856,13864,13872,13880,13888,13896,13904,13912,13920,13928,13936,13944,13952,13960,13968,13976,13984,13992,14000,14008,14016,14024,14032,14040,14048,14056,14064,14072,14080,14088,14096,14104,14112,14120,14128,14136,14144,14152,14160,14168,14176,14184,14192,14200,14208,14216,14224,14232,14240,14248,14256,14264,14272,14280,14288,14296,14304,14312,14320,14328,14336,14344,14352,14360,14368,14376,14384,14392,14400,14408,14416,14424,14432,14440,14448,14456,14464,14472,14480,14488,14496,14504,14512,14520,14528,14536,14544,14552,14560,14568,14576,14584,14592,14600,14608,14616,14624,14632,14640,14648,14656,14664,14672,14680,14688,14696,14704,14712,14720,14728,14736,14744,14752,14760,14768,14776,14784,14792,14800,14808,14816,14824,14832,14840,14848,14856,14864,14872,14880,14888,14896,14904,14912,14920,14928,14936,14944,14952,14960,14968,14976,14984,14992,15000,15008,15016,15024,15032,15040,15048,15056,15064,15072,15080,15088,15096,15104,15112,15120,15128,15136,15144,15152,15160,15168,15176,15184,15192,15200,15208,15216,15224,15232,15240,15248,15256,15264,15272,15280,15288,15296,15304,15312,15320,15328,15336,15344,15352,15360,15368,15376,15384,15392,15400,15408,15416,15424,15432,15440,15448,15456,15464,15472,15480,15488,15496,15504,15512,15520,15528,15536,15544,15552,15560,15568,15576,15584,15592,15600,15608,15616,15624,15632,15640,15648,15656,15664,15672,15680,15688,15696,15704,15712,15720,15728,15736,15744,15752,15760,15768,15776,15784,15792,15800,15808,15816,15824,15832,15840,15848,15856,15864,15872,15880,15888,15896,15904,15912,15920,15928,15936,15944,15952,15960,15968,15976,15984,15992,16000,16008,16016,16024,16032,16040,16048,16056,16064,16072,16080,16088,16096,16104,16112,16120,16128,16136,16144,16152,16160,16168,16176,16184,16192,16200,16208,16216,16224,16232,16240,16248,16256,16264,16272,16280,16288,16296,16304,16312,16320,16328,16336,16344,16352,16360,16368,16376,16384,16392,16400,16408,16416,16424,16432,16440,16448,16456,16464,16472,16480,16488,16496,16504,16512,16520,16528,16536,16544,16552,16560,16568,16576,16584,16592,16600,16608,16616,16624,16632,16640,16648,16656,16664,16672,16680,16688,16696,16704,16712,16720,16728,16736,16744,16752,16760,16768,16776,16784,16792,16800,16808,16816,16824,16832,16840,16848,16856,16864,16872,16880,16888,16896,16904,16912,16920,16928,16936,16944,16952,16960,16968,16976,16984,16992,17000,17008,17016,17024,17032,17040,17048,17056,17064,17072,17080,17088,17096,17104,17112,17120,17128,17136,17144,17152,17160,17168,17176,17184,17192,17200,17208,

1632,1624,1616,1608,1600,1592,1584,1576,1568,1560,1552,1544,1536,1528,1520,1512,1504,1496,1488,1480,1472,1464,1456,1448,1440,1432,1424,1416,1408,1400,1392,1384,1376,1368,1360,1352,1344,1336,1328,1320,1312,1304,1296,1288,1280,1272,1264,1256,1248,1240,1232,1224,1216,1208,1200,1192,1184,1176,1168,1160,1152,1144,1136,1128,1120,1112,1104,1096,1088,1080,1072,1064,1056,1048,1040,1032,1024,1016,1008,1000,992,984,976,968,960,952,944,936,928,920,912,904,896,888,880,872,864,856,848,840,832,824,816,808,800,792,784,776,768,760,752,744,736,728,720,712,704,696,688,680,672,664,656,648,640,632,624,616,608,600,592,584,576,568,560,552,544,536,528,520,512,504,496,488,480,472,464,456,448,440,432,424,416,408,400,392,384,376,368,360,352,344,336,328,320,312,304,296,288,280,272,264,256,248,240,232,224,216,208,200,192,184,176,168,160,152,144,136,128,120,112,104,96,88,80,72,64,56,48,40,32,24,16,8};

const uint32_t Piano_LUT[NS] =

{2304,2304,2304,2304,2304,2304,2305,2304,2306,2307,2305,2306,2307,1816,1775,1201,1412,1824,0,1294,1466,1822,2373,103,1711,1859,2298,2951,879,2403,2705,3011,3805,1467,2803,3030,3401,4095,1931,2870,3096,3317,4068,1885,2531,2613,2820,3441,1734,2091,2237,2187,2915,1493,1690,1805,1834,2371,1523,1587,1765,1764,2208,1621,1718,1930,1974,2188,2029,2047,2311,2326,2350,2384,2402,2643,2609,2362,2634,2664,2881,2805,2278,2632,2606,2868,2794,2066,2472,2424,2634,2675,1789,2234,2182,2461,2595,1680,2121,2127,2370,2629,1654,2064,2077,2336,2693,1795,2151,2141,2378,2795,1961,2236,2191,2418,2805,2119,2327,2258,2417,2783,2217,2372,2286,2326,2647,2263,2385,2301,2282,2491,2264,2360,2295,2214,2317,2242,2350,2288,2188,2157,2202,2309,2333,2238,2084,2200,2301,2361,2301,2061,2226,2305,2398,2359,2071,2256,2292,2405,2417,2078,2302,2306,2428,2497,2132,2366,2341,2458,2526,2146,2360,2290,2400,2491,2139,2359,2282,2343,2432,2144,2324,2254,2303,2367,2161,2305,2283,2321,2312,2181,2282,2308,2350,2313,2222,2293,2307,2330,2298,2249,2305,2340,2317,2272,2263,2303,2349,2332,2269,2283,2290,2344,2321,2267,2303,2305,2370,2345,2292,2329,2315,2389,2359,2286,2311,2293,2360,2325,2253,2270,2251,2340,2313,2225,2257,2235,2322,2323,2251,2268,2263,2333,2333,2266,2285,2284,2362,2351,2298,2301,2314,2361,2357,2292,2297,2297,2331,2352,2291,2292,2287,2317,2343,2302,2302,2262,2273,2316,2277,2303,2258,2269,2314,2280,2313,2272,2265,2306,2270,2313,2272,2279,2313,2306,2349,2313,2306,2324,2316,2358,2336,2333,2338,2334,2359,2340,2332,2330,2318,2325,2332,2322,2307,2306,2278,2292,2274,2280,2305,2267,2278,2263,2043,2176,2241,2287,2261,1991,2246,2287,2343,2419,2061,2317,2365,2422,2490,2221,2392,2447,2473,2547,2246,2371,2411,2469,2515,2292,2333,2352,2403,2461,2249,2263,2265,2303,2362,2239,2223,2212,2217,2276,2196,2199,2194,2194,2234,2228,2238,2245,2249,2241,2286,2307,2329,2304,2269,2346,2371,2379,2362,2272,2348,2389,2399,2377,2263,2342,2389,2408,2414,2251,2312,2353,2384,2414,2232,2272,2303,2336,2406,2224,2267,2286,2312,2384,2223,2270,2276,2300,2370,2230,2257,2267,2281,2347,2248,2285,2284,2296,2343,2263,2303,2301,2305,2321,2279,2316,2311,2314,2310,2293,2310,2312,2311,2291,2299,2298,2303,2314,2293,2316,2300,2295,2299,2278,2324,2300,2300,2289,2259,2329,2305,2306,2299,2255,2334,2314,2321,2317,2265,2340,2326,2338,2338,2276,2351,2341,2352,2357,2286,2342,2332,1985,1989,2379,1924,3300,997,2013,1871,2634,3776,1625,2530,2056,2720,2777,1411,1805,1800,2687,2475,2417,2176,2640,2899,2081,2470,1578,2372,2047,1827,2794,1989,2973,2207,2344,2918,2122,2416,1470,2015,2514,2470,2330,1946,2689,2838,2916,1869,1811,2245,2267,2653,1491,2206,2626,2752,2928,1800,2431,2337,2335,2161,1620,2278,2335,2557,2337,2469,2604,2527,2249,1865,2372,2050,2178,2055,2187,2833,2274,2556,2407,2294,2512,1905,2050,2035,2227,2455,2346,2403,2469,2551,2427,2432,1975,2106,2175,2164,2543,2102,2475,2473,2460,2692,2101,2205,2084,2154,2383,2153,2287,2375,2487,2586,2457,2175,2176,2158,2192,2283,2048,2327,2402,2439,2625,2262,2341,2252,2189,2309,2085,2175,2294,2287,2478,2439,2342,2428,2259,2318,2288,2074,2267,2171,2330,2481,2295,2455,2338,2357,1354,2060,2192,2373,3359,1494,2137,2579,2567,3662,1912,2307,2562,2565,3308,2067,2368,2603,2536,3289,2038,2239,2417,2359,2968,2070,2167,2262,2111,2724,1904,2062,2150,1973,2532,1946,2211,2219,2038,2420,1928,2186,2139,2091,2261,2080,2328,2304,2370,2317,2282,2423,2364,2490,2200,2353,2491,2432,2657,2190,2449,2518,2461,2657,2058,2390,2426,2404,2671,2003,2295,2311,2353,2598,1980,2202,2237,2289,2541,1967,2078,2135,2202,2478,1994,2095,2175,2242,2585,2064,2135,2208,2243,2630,2105,2254,2244,2306,2700,2213,2376,2284,2341,2686,2228,2421,2285,2323,2635,2292,2499,2360,2336,2491,2267,2456,2268,2267,2296,2183,2335,2206,2282,2195,2239,2326,2231,2288,2078,2201,2220,2220,2295,2045,2225,2267,2323,2409,2136,2285,2305,2351,2460,2163,2282,2338,2384,2565,2239,2302,2370,2318,2665,2239,23

92,2445,1967,2529,2042,2443,2057,2326,2238,2664,2000,2588,1772,2693,1995,2301,2047,2116,2488,2141,2538,2047,2549,2094,2555,2067,2459,2041,2471,2143,2615,2182,2535,2284,2372,2501,2156,2610,1893,2543,2015,2689,2125,2518,2035,2555,2264,2421,2141,2093,2410,2162,2574,1974,2442,2093,2639,2110,2451,1931,2488,2151,2470,2200,2242,2464,2180,2626,2055,2523,2038,2516,2255,2362,2229,2329,2313,2435,2295,2322,2252,2297,2321,2342,1605,1571,1786,2621,3998,2445,2565,1758,2851,1242,2548,2628,2490,2053,1843,2257,2454,3615,1971,2316,1869,2078,1547,2458,2766,2583,2819,1734,2152,2240,3039,1890,2403,1992,2040,2116,2359,2666,2621,3044,1677,2122,2125,2573,2094,2491,2177,2181,2473,2154,2638,2568,2912,1676,2062,1944,2314,2406,2529,2362,2338,2551,1918,2223,2632,3063,2147,2168,1529,1998,2734,2866,3018,2381,2243,1371,2069,2442,2783,2527,2318,1825,2027,2537,2411,2859,2540,2299,1413,1872,2046,2626,2974,2417,2144,1914,2326,2096,2736,2571,2319,1789,1811,1953,2556,3061,2548,2378,2001,2087,1937,2650,2426,2553,2271,1977,2053,2360,2942,2476,2570,2069,1836,1809,2406,2503,2756,2482,1995,2029,2244,2728,2300,2581,2038,1954,2041,2177,2475,2661,2614,2192,2230,2207,2348,2162,2451,2147,2309,2308,2258,2379,2466,2536,2161,2295,2162,2261,2249,2348,2289,2419,2393,2254,2304,2374,2406,2207,2311,2148,2278};

1,2439,1535,1088,2938,1572,1790,1799,1184,2104,1644,2365,1603,1064,2577,1095,2637,1901,805,2660,1020,280
0,1540,1405,2319,1432,2979,1455,1631,2330,1398,2968,1757,1433,2205,1318,2629,1653,1706,1973,1024,2741,14
31,1981,1879,1036,2559,1484,2477,1399,1366,2354,1492,2766,1531,1362,2266,1612,2783,1630,1484,2162,1404,2
721,1662,1608,2055,1136,2780,1455,1949,1801,1137,2651,1436,2323,1599,1282,2439,1689,2486,1617,1389,2243,
1669,2657,1685,1448,2166,1517,2693,1622,1618,1995,1260,2789,1320,1931,1809,1281,2636,1452,2104,1707,134
7,2441,1566,2342,1659,1399,2420,1582,2507,1715,1477,2264,1508,2653,1572,1554,2179,1349,2672,1450,1740,19
36,1311,2598,1413,1974,1756,1349,2533,1494,2209,1755,1450,2355,1672,2333,1729,1466,2334,1573,2526,1669,1
488,2262,1413,2674,1420,1728,1970,1369,2601,1404,1931,1789,1463,2461,1543,2039,1799,1433,2419,1608,2208,
1826,1372,2474,1540,2483,1643,1507,2267,1454,2603,1525,1657,1970,1463,2594,1449,1773,1980,1527,2262,220
7,1649,1533,1517,2729,1376,2539,1564,1255,2376,1869,2341,1847,1378,2126,1587,2683,1686,1565,2013,1310,26
87,1515,1994,1632,1588,2272,1823,1709,2078,1175,2469,1621,2210,1632,1606,2119,1759,2206,1891,1284,2543,1
300,2695,1421,1770,1787,1772,2229,1803,1535,2057,1381,2561,1599,1925,1881,1503,2404,1655,2097,1827,1542,
2266,1780,2132,1983,1136,2322,1413,2726,1938,1007,2296,2049,1817,2430,2660,1003,1626,2383,1345,1113,1224
,2722,1895,43,163,2135,1379,1897,720,1878,1038,1379,1212,834,3011,2250,2142,2307,2041,3555,2238,1966,203
0,1832,3300,1576,1839,1536,1532,2969,1426,1663,1851,1374,3150,1204,2241,1610,1035,2082,1242,1901,2508,67
8,2097,713,1845,1880,1255,1902,2163,2342,2907,1745,2179,2095,2458,2033,1943,1423,2156,2199,1943,2173};
const uint32_t Drum_LUT[NS] =
{2047,2047,2047,2047,2048,2046,2045,2042,2050,2050,2048,2049,2042,1855,2239,2632,2118,669,4090,68,1563,4
091,0,2681,4088,798,1275,2260,371,1433,1341,1523,2104,3611,4058,3881,2630,2188,1702,2,0,927,2330,2509,256
2,3313,3252,3330,3295,1541,484,1080,1712,2285,2368,1778,2069,2174,2108,2405,2263,2158,2246,2033,1457,107
3,1579,1962,2241,2694,2703,2369,2517,2493,1943,1592,1375,1355,1692,1928,2195,2621,2638,2306,2072,1890,18
45,1930,2061,2153,1979,1867,1889,1872,1957,2030,2169,2449,2513,2409,2147,1719,1495,1476,1741,2015,2296,2
411,2434,2368,2304,1928,1715,1679,1773,2038,2218,2249,2193,2062,1947,1887,1881,2054,2242,2291,2214,2095,
1934,1822,1801,1876,1987,2141,2240,2273,2226,2105,2026,1941,1870,1868,1972,2082,2157,2168,2116,2039,195
7,1950,2011,2076,2124,2133,2099,2049,1993,1953,1957,1998,2065,2118,2111,2079,2072,2060,2032,2010,1982,19
96,2021,2042,2064,2099,2122,2116,2084,2027,1984,1976,1987,2018,2048,2075,2087,2079,2059,2051,2036,2032,2
046,2062,2103,2051,1807,1977,1722,2488,2530,2643,0,4084,1922,8,4091,3087,603,2797,872,479,1843,1756,1827,
2930,3892,4095,3287,2176,2328,80,0,6,1711,2655,2406,2814,3400,3212,3500,2669,762,655,1442,1937,2345,2046,
1876,2215,2084,2256,2417,2193,2176,2194,1811,1172,1261,1808,2031,2473,2808,2531,2394,2570,2242,1741,1483
,1303,1488,1829,2003,2410,2715,2494,2185,1978,1833,1894,1979,2147,2091,1890,1869,1881,1893,1997,2089,231
2,2468,2394,2353,1900,1661,1444,1620,1874,2135,2361,2467,2387,2364,2129,1814,1688,1683,1863,2155,2260,22
27,2136,1984,1908,1880,1939,2162,2283,2264,2161,2005,1871,1808,1815,1927,2060,2201,2267,2260,2181,2062,1
978,1904,1854,1906,2020,2127,2175,2149,2086,2001,1941,1970,2038,2105,2134,2119,2085,2023,1973,1958,1985,
2032,2096,2123,2096,2074,2069,2049,2021,1995,1987,2009,2030,2048,2081,2108,2123,2103,2060,2006,1976,198
0,2001,2031,2057,2082,2088,2074,2053,2041,2027,2036,2061,2085,2063,1995,816,4095,4,3874,3787,8,1005,4077,
11,1367,3566,1468,1867,2277,2853,2400,2518,2036,2130,1397,1576,2414,2183,1809,958,1214,2614,2431,1240,25
58,2495,2208,1721,2755,1794,2090,2208,1757,1884,2422,1898,1613,2144,2261,1738,2018,2132,1648,2013,2721,1
837,1986,2410,1888,2101,2324,1887,1999,2256,1909,1981,2245,2093,1916,2122,2046,1910,2075,2113,1931,2054,
2093,1960,2023,2109,2053,2010,2066,2072,2014,2078,2078,2007,2099,2131,2044,2020,2059,2014,1983,2039,201
5,2020,2095,2083,2059,2075,2111,2051,2038,2050,2006,2033,2050,2063,2018,2035,2080,2028,2054,2057,2072,20
44,2091,2036,2067,2068,2022,2019,2046,2040,2039,2051,2062,2032,2048,2016,2032,2043,2066,2046,2044,2054,2
040,2042,2048,2024,2045,2051,2051,2043,2045,2041,2049,2056,2053,2043,2064,2049,2040,2048,2053,2045,2046,
2047,2037,2044,2041,2047,2046,2052,2040,2038,2051,2049,2041,2035,2045,2051,2044,2027,2050,2045,2036,204
4,2055,2058,2049,2039,2079,1470,3823,3,1105,4095,0,2316,4086,1094,1450,1558,407,2020,1499,1433,2500,3569,
3878,4004,2489,2171,1287,1,0,1194,2801,2616,2572,3439,2968,3505,3366,1082,46,973,1501,2167,2520,1935,219
3,2342,2458,2293,2321,2064,2103,1924,1321,911,1698,2029,2730,3267,2575,2168,2419,2179,1722,1399,1374,147
6,1856,1998,2165,2585,2714,2434,2231,1783,1733,1843,2095,2076,1840,1691,1693,2041,2459,2510,2431,2477,21

```

42,2034,1660,1361,1524,1804,2297,2453,2464,2287,2025,1889,1830,1986,2000,2085,2142,2180,2077,2066,1859,1
714,1736,1969,2245,2455,2553,2248,1978,1796,1644,1803,1900,2108,2225,2414,2260,2007,1906,1814,1862,1946,
2144,2126,2103,2211,2132,2080,1832,1859,1982,2110,2070,2095,2144,2176,2046,1873,1915,1979,2035,2132,2063
,2082,2077,1991,1927,1969,2082,2025,1999,2038,2109,2135,2040,2071,2080,1996,1929,1984,2028,2071,2044,210
0,2082,2152,2043,2022,2015,1989,1985,2059,2013,2075,2057,2067,2022,2039,2090,2168,2076,2343,1732,2105,22
58,1785,2278,1974,2132,1869,2198,2260,1694,2030,2007,2075,1798,1912,2186,1991,1897,2093,2179,2192,2199,2
259,1809,2023,2091,2032,2090,2054,1969,2059,2091,2049,2151,1906,2085,1958,2041,2080,2018,2117,2150,2018,
2043,1998,2031,2096,1984,2044,2075,2022,2092,2017,2039,2090,2015,2050,2040,2046,2059,2051,2045,2075,203
0,2039,2036,2017,2058,2048,2062,2082,2054,2057,2040,2033,2040,2046,2053,2049,2036,2053,2035,2034,2046,19
88,2073,1926,2045,2063,2030,2090,2043,2050,2045,2054,2035,2101,1989,2079,2071,2019,2050,2046,2063,2026,2
040,2033,2065,2039,2054,2071,2047,2063,2040,2037,2053,2043,2041,2056,2027,2058,2032,2033,2047,2047,2059,
2067,2063,2069,2052,2044,2040,2031,2034,2030,2039,2041,2052,2049,2054,2053,2054,2053,2051,2045,2051,204
7,2042,2043,2043,2041,2041,2048,2047,2052,2049,2046,2051,2045,2048,2047,2049,2048,2048,2042,2042,2041,20
42,2044,2047,2049,2056,2051,2054,2044,2043,2046,2041,2057,2329,2065,1811,2299,2161,1975,2572,1944,2470,2
129,1472,1954,2326,2014,1834,2163,1761,2364,1901,2346,2183,2079,1691,2257,1773,1728,2093,1970,1995,1983,
1976,2091,1921,2026,2245,1971,1926,2158,2052,2128,1909,2038,2008,2006,2077,1990,2021,2105,2013,2029,211
2,2066,2101,2075,1974,2110,2040,1972,2129,2087,2096,2133,2058,2041,2040,2046,2084,2059,2037,2062,2028,20
56,2029,2053,2046,2050,2044,2058,2047,2054,2042,2039,2042,2047,2042,2047,2052,2047,2039,2113,2040,2069,2
048,2076,2053,2030,2036,2061,2039,2043,1999,2068,2052,2044,2035,2033};

```

```
uint32_t prev_tick = 0;
```

```
uint32_t waveform = 0;
```

```
// TODO: Equation to calculate TIM2_Ticks
```

```
uint32_t TIM2_Ticks = TIM2CLK/(F_SIGNAL*NS); // How often to write new LUT value
```

```
uint32_t DestAddress = (uint32_t) &(TIM3->CCR3); // Write LUT TO TIM3->CCR3 to modify PWM duty cycle
```

```
/* USER CODE END PV */
```

```
/* Private function prototypes -----*/
```

```
void SystemClock_Config(void);
```

```
static void MX_GPIO_Init(void);
```

```
static void MX_DMA_Init(void);
```

```
static void MX_TIM2_Init(void);
```

```
static void MX_TIM3_Init(void);
```

```
/USER CODE BEGIN PFP /
```

```
void EXTI0_IRQHandler(void);
```

```
/USER CODE END PFP */
```

```
/* Private user code -----*/
```

```
/USER CODE BEGIN 0 */
```

```
/* USER CODE END 0 */
```

```
/**
```

- @brief The application entry point.

- @retval int

```
*/
```

```
int main(void)
```

```
{
```



```

/* USER CODE BEGIN 1 */

/* USER CODE END 1 */

/* MCU Configuration-----*/

/* Reset of all peripherals, Initializes the Flash interface and the Systick. */
HAL_Init();

/* USER CODE BEGIN Init */

/* USER CODE END Init */

/* Configure the system clock */
SystemClock_Config();

/* USER CODE BEGIN SysInit */

/* USER CODE END SysInit */

/* Initialize all configured peripherals */
MX_GPIO_Init();
MX_DMA_Init();
MX_TIM2_Init();
MX_TIM3_Init();
/USER CODE BEGIN 2 /
// TODO: Start TIM3 in PWM mode on channel 3;
HAL_TIM_PWM_Start(&htim3, TIM_CHANNEL_3);
TIM3->CCR3 = (1<<15);
// TODO: Start TIM2 in Output Compare (OC) mode on channel 1
HAL_TIM_OC_Start(&htim2, TIM_CHANNEL_1);
// TODO: Start DMA in IT mode on TIM2->CH1. Source is LUT and Dest is TIM3->CCR3; start with Sine LUT
HAL_DMA_Start_IT(&hdma_tim2_ch1, Sin_LUT, DestAddress, NS);
// TODO: Write current waveform to LCD(Sine is the first waveform)
init_LCD();
lcd_command(CLEAR);
lcd_update_line("Sine");
// TODO: Enable DMA (start transfer from LUT to CCR)
__HAL_TIM_ENABLE_DMA(&htim2, TIM_DMA_CC1);
/USER CODE END 2 */

/* Infinite loop */
/USER CODE BEGIN WHILE /
while (1)
{
/USER CODE END WHILE */

/* USER CODE BEGIN 3 */

}
/* USER CODE END 3 */

```

```
}
```

```
/**
```

- @brief System Clock Configuration
- @retval None

```
*/
```

```
void SystemClock_Config(void)
```

```
{
```

```
RCC_OscInitTypeDef RCC_OscInitStruct = {0};
```

```
RCC_ClkInitTypeDef RCC_ClkInitStruct = {0};
```

```
/** Configure the main internal regulator output voltage
```

```
*/
```

```
__HAL_RCC_PWR_CLK_ENABLE();
```

```
__HAL_PWR_VOLTAGESCALING_CONFIG(PWR_REGULATOR_VOLTAGE_SCALE3);
```

```
/** Initializes the RCC Oscillators according to the specified parameters
```

- in the RCC_OscInitTypeDef structure.

```
*/
```

```
RCC_OscInitStruct.OscillatorType = RCC_OSCILLATORTYPE_HSI;
```

```
RCC_OscInitStruct.HSIState = RCC_HSI_ON;
```

```
RCC_OscInitStruct.HSICalibrationValue = RCC_HSICALIBRATION_DEFAULT;
```

```
RCC_OscInitStruct.PLL.PLLState = RCC_PLL_NONE;
```

```
if (HAL_RCC_OscConfig(&RCC_OscInitStruct) != HAL_OK)
```

```
{
```

```
Error_Handler();
```

```
}
```

```
/** Initializes the CPU, AHB and APB buses clocks
```

```
*/
```

```
RCC_ClkInitStruct.ClockType = RCC_CLOCKTYPE_HCLK|RCC_CLOCKTYPE_SYSCLK
```

```
|RCC_CLOCKTYPE_PCLK1|RCC_CLOCKTYPE_PCLK2;
```

```
RCC_ClkInitStruct.SYSCLKSource = RCC_SYSCLKSOURCE_HSI;
```

```
RCC_ClkInitStruct.AHBCLKDivider = RCC_SYSCLK_DIV1;
```

```
RCC_ClkInitStruct.APB1CLKDivider = RCC_HCLK_DIV1;
```

```
RCC_ClkInitStruct.APB2CLKDivider = RCC_HCLK_DIV1;
```

```
if (HAL_RCC_ClockConfig(&RCC_ClkInitStruct, FLASH_LATENCY_0) != HAL_OK)
```

```
{
```

```
Error_Handler();
```

```
}
```

```
}
```

```
/**
```

- @brief TIM2 Initialization Function
- @param None

- @retval None

```

*/
static void MX_TIM2_Init(void)
{

/* USER CODE BEGIN TIM2_Init 0 */

/* USER CODE END TIM2_Init 0 */

TIM_ClockConfigTypeDef sClockSourceConfig = {0};
TIM_MasterConfigTypeDef sMasterConfig = {0};
TIM_OC_InitTypeDef sConfigOC = {0};

/* USER CODE BEGIN TIM2_Init 1 */

/* USER CODE END TIM2_Init 1 */
htim2.Instance = TIM2;
htim2.Init.Prescaler = 0;
htim2.Init.CounterMode = TIM_COUNTERMODE_UP;
// htim2.Init.Period = 4294967295;
htim2.Init.Period = TIM2_Ticks;//#
htim2.Init.ClockDivision = TIM_CLOCKDIVISION_DIV1;
htim2.Init.AutoReloadPreload = TIM_AUTORELOAD_PRELOAD_DISABLE;
if (HAL_TIM_Base_Init(&htim2) != HAL_OK)
{
Error_Handler();
}
sClockSourceConfig.ClockSource = TIM_CLOCKSOURCE_INTERNAL;
if (HAL_TIM_ConfigClockSource(&htim2, &sClockSourceConfig) != HAL_OK)
{
Error_Handler();
}
if (HAL_TIM_OC_Init(&htim2) != HAL_OK)
{
Error_Handler();
}
sMasterConfig.MasterOutputTrigger = TIM_TRGO_RESET;
sMasterConfig.MasterSlaveMode = TIM_MASTERSLAVEMODE_DISABLE;
if (HAL_TIMEx_MasterConfigSynchronization(&htim2, &sMasterConfig) != HAL_OK)
{
Error_Handler();
}
sConfigOC.OCMode = TIM_OCMODE_TIMING;
sConfigOC.Pulse = 0;
sConfigOC.OCpolarity = TIM_OCPOLARITY_HIGH;
sConfigOC.OCFastMode = TIM_OCFAST_DISABLE;
if (HAL_TIM_OC_ConfigChannel(&htim2, &sConfigOC, TIM_CHANNEL_1) != HAL_OK)
{
Error_Handler();
}

```

```

}
/USER CODE BEGIN TIM2_Init 2 /
/TIM2_CH1 DMA Init */
__HAL_RCC_DMA1_CLK_ENABLE();

hdma_tim2_ch1.Instance = DMA1_Stream5;
hdma_tim2_ch1.Init.Channel = DMA_CHANNEL_3; // TIM2_CH1 is on channel 3
hdma_tim2_ch1.Init.Direction = DMA_MEMORY_TO_PERIPH; // Memory -> TIM3->CCR3
hdma_tim2_ch1.Init.PeriphInc = DMA_PINC_DISABLE; // Peripheral address fixed
hdma_tim2_ch1.Init.MemInc = DMA_MINC_ENABLE; // Memory address increments
hdma_tim2_ch1.Init.PeriphDataAlignment = DMA_PDATAALIGN_WORD;
hdma_tim2_ch1.Init.MemDataAlignment = DMA_MDATAALIGN_WORD;
hdma_tim2_ch1.Init.Mode = DMA_CIRCULAR; // Repeat LUT automatically
hdma_tim2_ch1.Init.Priority = DMA_PRIORITY_HIGH;
hdma_tim2_ch1.Init.FIFOMode = DMA_FIFOMODE_DISABLE;

if (HAL_DMA_Init(&hdma_tim2_ch1) != HAL_OK)
{
    Error_Handler();
}

/* Link DMA handle to TIM2 handle /
__HAL_LINKDMA(&htim2, hdma[TIM_DMA_ID_CC1], hdma_tim2_ch1);
/USER CODE END TIM2_Init 2 */

}

/**
 *
 * @brief TIM3 Initialization Function
 * @param None
 * @retval None
 */
static void MX_TIM3_Init(void)
{
    /USER CODE BEGIN TIM3_Init 0 */

    /USER CODE END TIM3_Init 0 */

    TIM_ClockConfigTypeDef sClockSourceConfig = {0};
    TIM_MasterConfigTypeDef sMasterConfig = {0};
    TIM_OC_InitTypeDef sConfigOC = {0};

    /USER CODE BEGIN TIM3_Init 1 */

    /USER CODE END TIM3_Init 1 /
    htim3.Instance = TIM3;
    htim3.Init.Prescaler = 0;
    htim3.Init.CounterMode = TIM_COUNTERMODE_UP;
    // htim3.Init.Period = 65535;

```

```

htim3.Init.Period = (1<<12)-1;///  

htim3.Init.ClockDivision = TIM_CLOCKDIVISION_DIV1;  

htim3.Init.AutoReloadPreload = TIM_AUTORELOAD_PRELOAD_DISABLE;  

if (HAL_TIM_Base_Init(&htim3) != HAL_OK)  
{  
    Error_Handler();  
}  

sClockSourceConfig.ClockSource = TIM_CLOCKSOURCE_INTERNAL;  

if (HAL_TIM_ConfigClockSource(&htim3, &sClockSourceConfig) != HAL_OK)  
{  
    Error_Handler();  
}  

if (HAL_TIM_PWM_Init(&htim3) != HAL_OK)  
{  
    Error_Handler();  
}  

sMasterConfig.MasterOutputTrigger = TIM_TRGO_RESET;  

sMasterConfig.MasterSlaveMode = TIM_MASTERSLAVEMODE_DISABLE;  

if (HAL_TIMEx_MasterConfigSynchronization(&htim3, &sMasterConfig) != HAL_OK)  
{  
    Error_Handler();  
}  

sConfigOC.OCMode = TIM_OCMODE_PWM1;  

sConfigOC.Pulse = 0;  

// sConfigOC.Pulse = 1;///  

sConfigOC.OCpolarity = TIM_OCPOLARITY_HIGH;  

sConfigOC.OCFastMode = TIM_OCFAST_DISABLE;  

if (HAL_TIM_PWM_ConfigChannel(&htim3, &sConfigOC, TIM_CHANNEL_3) != HAL_OK)  
{  
    Error_Handler();  
}  

/ USER CODE BEGIN TIM3_Init 2 */  
  

/* USER CODE END TIM3_Init 2 */  

HAL_TIM_MspPostInit(&htim3);  
  

}  
  

/**  

 * Enable DMA controller clock  

 */  

static void MX_DMA_Init(void)  
{  
  

/* DMA controller clock enable */  

__HAL_RCC_DMA1_CLK_ENABLE();

```

```

/* DMA interrupt init /
/DMA1_Stream5_IRQn interrupt configuration */
HAL_NVIC_SetPriority(DMA1_Stream5_IRQn, 0, 0);
HAL_NVIC_EnableIRQ(DMA1_Stream5_IRQn);

}

/**
 *
 * @brief GPIO Initialization Function
 * @param None
 * @retval None
 */
static void MX_GPIO_Init(void)
{
    GPIO_InitTypeDef GPIO_InitStruct = {0};
    /USER CODE BEGIN MX_GPIO_Init_1 */

/* USER CODE END MX_GPIO_Init_1 */

/* GPIO Ports Clock Enable */
__HAL_RCC_GPIOH_CLK_ENABLE();
__HAL_RCC_GPIOA_CLK_ENABLE();
__HAL_RCC_GPIOB_CLK_ENABLE();

// -----
// LCD pins configuration
// -----
// Configure PC14 (RS) and PC15 (E) as output push-pull
GPIO_InitStruct.Pin = GPIO_PIN_14 | GPIO_PIN_15;
GPIO_InitStruct.Mode = GPIO_MODE_OUTPUT_PP;
GPIO_InitStruct.Pull = GPIO_NOPULL;
GPIO_InitStruct.Speed = GPIO_SPEED_FREQ_LOW;
HAL_GPIO_Init(GPIOC, &GPIO_InitStruct);

// Configure PB8 (D4) and PB9 (D5) as output push-pull
GPIO_InitStruct.Pin = GPIO_PIN_8 | GPIO_PIN_9;
HAL_GPIO_Init(GPIOB, &GPIO_InitStruct);

// Configure PA12 (D6) and PA15 (D7) as output push-pull
GPIO_InitStruct.Pin = GPIO_PIN_12 | GPIO_PIN_15;
HAL_GPIO_Init(GPIOA, &GPIO_InitStruct);

// Set all LCD pins LOW initially
HAL_GPIO_WritePin(GPIOC, GPIO_PIN_14 | GPIO_PIN_15, GPIO_PIN_RESET);
HAL_GPIO_WritePin(GPIOB, GPIO_PIN_8 | GPIO_PIN_9, GPIO_PIN_RESET);
HAL_GPIO_WritePin(GPIOA, GPIO_PIN_12 | GPIO_PIN_15, GPIO_PIN_RESET);

// -----
// Button0 configuration (PA0)
// -----

```

```

GPIO_InitStruct.Pin = Button0_Pin;
GPIO_InitStruct.Mode = GPIO_MODE_IT_RISING; // Interrupt on rising edge
GPIO_InitStruct.Pull = GPIO_PULLUP; // Use pull-up resistor
HAL_GPIO_Init(GPIOA, &GPIO_InitStruct);

// Enable and set EXTI line 0 interrupt priority
HAL_NVIC_SetPriority(EXTI0_IRQn, 2, 0);
HAL_NVIC_EnableIRQ(EXTI0_IRQn);

/* USER CODE END MX_GPIO_Init_2 */

}

/* USER CODE BEGIN 4 */
void EXTI0_IRQHandler(void){

// TODO: Debounce using HAL_GetTick()
if (HAL_GetTick()-prev_tick>500) { //if more than 0.5 seconds have passed
prev_tick = HAL_GetTick();

// TODO: Disable DMA transfer and abort IT, then start DMA in IT mode with new LUT and re-enable transfer
// HINT: Consider using C's "switch" function to handle LUT changes
__HAL_TIM_DISABLE_DMA(&htim2,TIM_DMA_CC1);
HAL_DMA_Abort_IT(&hdma_tim2_ch1);
waveform = (waveform+1)%6;
switch (waveform) {
case 0:
HAL_DMA_Start_IT(&hdma_tim2_ch1, Sin_LUT, DestAddress, NS);
Lcd_update_line("Sine");
break;
case 1:
HAL_DMA_Start_IT(&hdma_tim2_ch1, Saw_LUT, DestAddress, NS);
Lcd_update_line("Sawtooth");
break;
case 2:
HAL_DMA_Start_IT(&hdma_tim2_ch1, Triangle_LUT, DestAddress, NS);
Lcd_update_line("Triangular");
break;
case 3:
HAL_DMA_Start_IT(&hdma_tim2_ch1, Piano_LUT, DestAddress, NS);
Lcd_update_line("Piano");
break;
case 4:
HAL_DMA_Start_IT(&hdma_tim2_ch1, Guitar_LUT, DestAddress, NS);
Lcd_update_line("Guitar");
break;
case 5:
HAL_DMA_Start_IT(&hdma_tim2_ch1, Drum_LUT, DestAddress, NS);
Lcd_update_line("Drum");

```

```

break;
}
__HAL_TIM_ENABLE_DMA(&htim2, TIM_DMA_CC1);
}

HAL_GPIO_EXTI_IRQHandler(Button0_Pin); // Clear interrupt flags
}
/* USER CODE END 4 */

/**

```

- @brief This function is executed in case of error occurrence.
- @retval None

```

/
void Error_Handler(void)
{
/USER CODE BEGIN Error_Handler_Debug /
/ User can add his own implementation to report the HAL error return state /
__disable_irq();
while (1)
{
}
/USER CODE END Error_Handler_Debug /
}
#ifdef USE_FULL_ASSERT
/*

```

- @brief Reports the name of the source file and the source line number
- where the assert_param error has occurred.

- @param file: pointer to the source file name
- @param line: assert_param error line source number
- @retval None

```

*/
void assert_failed(uint8_t file, uint32_t line)
{
/USER CODE BEGIN 6 /
/ User can add his own implementation to report the file name and line number,
ex: printf("Wrong parameters value: file %s on line %d\r\n", file, line) /
/USER CODE END 6 /
}
#endif /USE_FULL_ASSERT */

```